Portfolio Project: Predicting Breast Cancer Malignancy

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```r
library(tidyverse)
library(tidymodels)
library(glmnet)
library(broom)
library(pROC)
library(yardstick)
library(forcats)

theme_set(theme_minimal())

set.seed(123)
```

1 Abstract

This project evaluates whether tumor morphology measurements can be used to predict breast cancer malignancy using a lasso-penalized logistic regression model. Using the Wisconsin Diagnostic Breast Cancer dataset, I compare a regularized classifier against an unpenalized logistic regression baseline. The analysis emphasizes predictive performance, model parsimony, and interpretability. Results suggest that features related to tumor size and boundary irregularity are the strongest predictors of malignancy, while regularization improves interpretability by shrinking redundant predictors to zero.

2 Research Question

Can morphologic tumor features derived from digitized breast mass images be used to accurately classify tumors as benign or malignant, and which features contribute most strongly to prediction?

3 Data and Preprocessing

```r
raw_df <- read_csv("data.csv", show_col_types = FALSE)
```

```
## New names:
## * `` -> `...33`
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```r
df <- raw_df %>%
  select(where(~ !all(is.na(.)))) %>%
  mutate(
    diagnosis = factor(diagnosis, levels = c("B", "M"))
  )
```

```r
cat("Observations:", nrow(df), "\n")
```

```
## Observations: 568
```

```r
cat("Predictors:", ncol(df) - 2, "\n")
```

```
## Predictors: 30
```

```r
cat("Malignant:", sum(df$diagnosis == "M"), "\n")
```

```
## Malignant: 212
```

```r
cat("Benign:", sum(df$diagnosis == "B"), "\n")
```

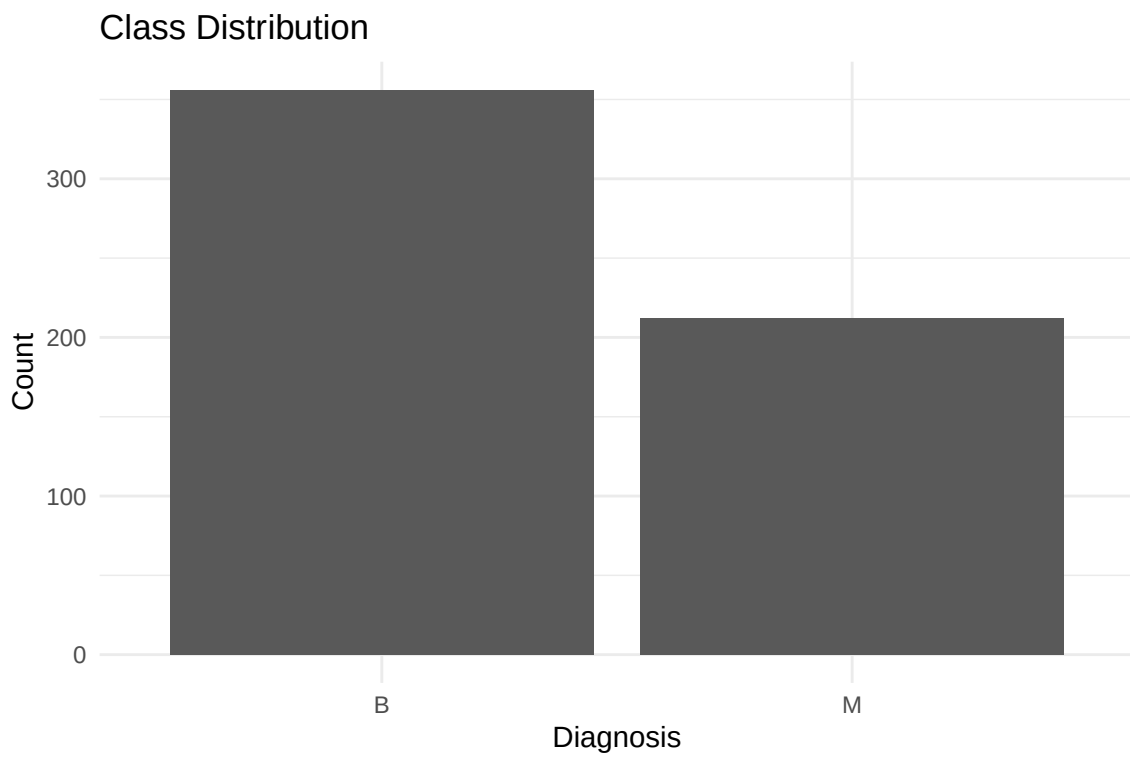
```
## Benign: 356
```

The dataset contains 569 observations and 30 predictor variables after excluding the ID field and removing any empty columns. The binary outcome is `diagnosis`, where `B` denotes benign and `M` denotes malignant.

4 Exploratory Data Analysis

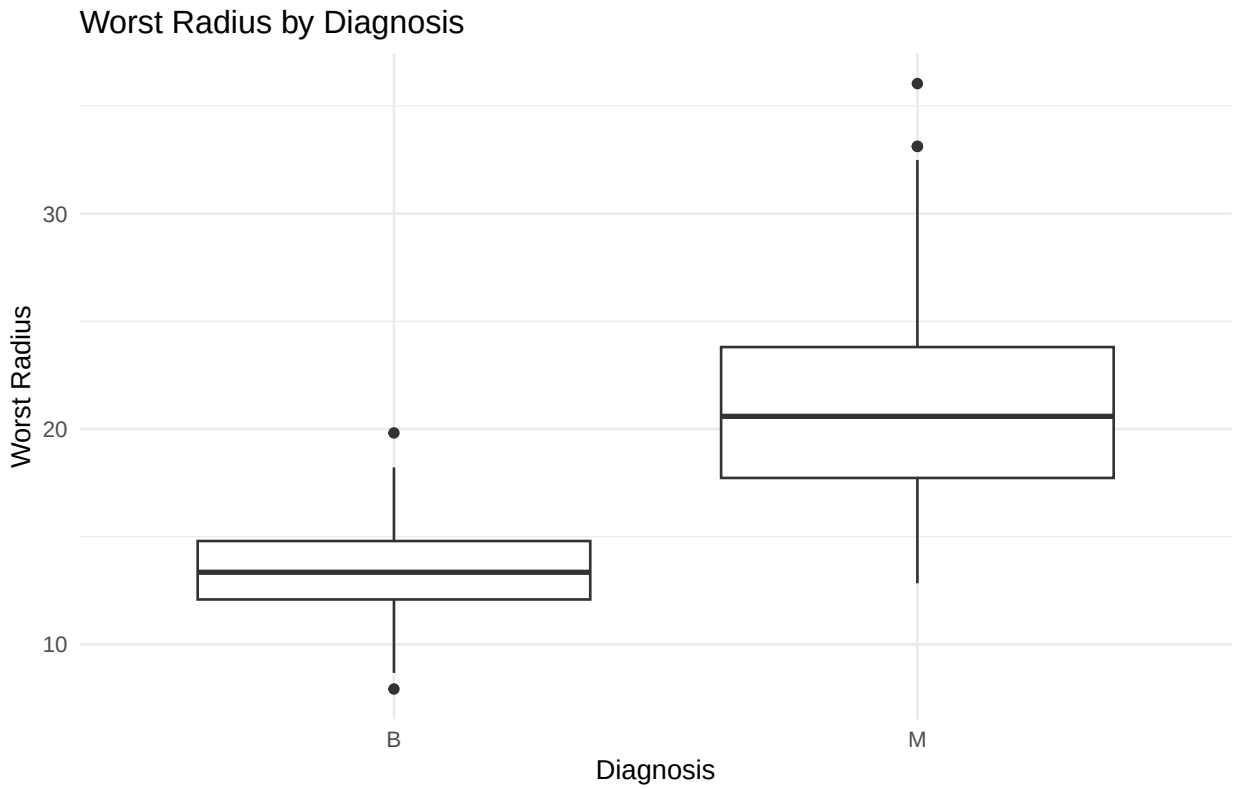
4.1 Outcome balance

```r
df %>%
  count(diagnosis) %>%
  ggplot(aes(x = diagnosis, y = n)) +
  geom_col() +
  labs(
    title = "Class Distribution",
    x = "Diagnosis",
    y = "Count"
  )
```



4.2 Example feature separation

```
df %>%
  ggplot(aes(x = diagnosis, y = radius_worst)) +
  geom_boxplot() +
  labs(
    title = "Worst Radius by Diagnosis",
    x = "Diagnosis",
    y = "Worst Radius"
  )
```



These plots provide an initial indication that malignant tumors tend to have larger and more irregular morphologic measurements than benign tumors.

5 Train/Test Split and Resampling

```
model_df <- df %>% select(-id)

split_obj <- initial_split(model_df, prop = 0.80, strata = diagnosis)
train_df  <- training(split_obj)
test_df   <- testing(split_obj)

cv_folds <- vfold_cv(train_df, v = 5, strata = diagnosis)
```

I use an 80/20 stratified train/test split. Model tuning is conducted using 5-fold cross-validation on the training data only.

6 Modeling Strategy

The main model is a lasso logistic regression. Lasso is appropriate here because many predictors are correlated, and L1 regularization can improve parsimony by shrinking weaker coefficients to zero.

6.1 Recipe

```
base_recipe <- recipe(diagnosis ~ ., data = train_df) %>%
  step_normalize(all_numeric_predictors())
```

6.2 Lasso logistic regression

```
lasso_spec <- logistic_reg(
  penalty = tune(),
  mixture = 1
) %>%
  set_engine("glmnet")

lasso_workflow <- workflow() %>%
  add_recipe(base_recipe) %>%
  add_model(lasso_spec)
```

6.3 Tune penalty by cross-validation

```
penalty_grid <- grid_regular(
  penalty(range = c(-4, 1)),
  levels = 40
)

lasso_tuned <- tune_grid(
  lasso_workflow,
  resamples = cv_folds,
  grid = penalty_grid,
  metrics = metric_set(roc_auc, accuracy, sens, spec)
)

best_penalty <- select_best(lasso_tuned, metric = "roc_auc")
best_penalty
```

```
## # A tibble: 1 x 2
##   penalty .config
##     <dbl> <chr>
## 1 0.00191 pre0_mod11_post0
```

7 Final Lasso Model

```
final_lasso_wf <- finalize_workflow(lasso_workflow, best_penalty)
final_lasso_fit <- fit(final_lasso_wf, data = train_df)
```

7.1 Test-set predictions

```
lasso_pred <- predict(final_lasso_fit, test_df, type = "prob") %>%
  bind_cols(predict(final_lasso_fit, test_df)) %>%
  bind_cols(test_df %>% select(diagnosis))
```

7.2 Performance metrics

```
lasso_class_metrics <- metric_set(accuracy, sens, spec)(
  lasso_pred,
  truth = diagnosis,
  estimate = .pred_class
)

lasso_auc <- roc_auc(lasso_pred, truth = diagnosis, .pred_M)

lasso_class_metrics
```

```
## # A tibble: 3 x 3
##   .metric  .estimator .estimate
##   <chr>    <chr>          <dbl>
## 1 accuracy binary         0.974
## 2 sens     binary         0.958
## 3 spec     binary         1
```

```
lasso_auc
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 roc_auc binary      0.000323
```

7.3 Confusion matrix

```
conf_mat(lasso_pred, truth = diagnosis, estimate = .pred_class)
```

```
##           Truth
## Prediction  B  M
##          B 69  0
##          M  3 43
```

8 Baseline Comparison: Standard Logistic Regression

To contextualize the lasso model, I compare it with an unpenalized logistic regression baseline fit using the same preprocessing recipe.

```r
glm_spec <- logistic_reg() %>%
  set_engine("glm")

glm_workflow <- workflow() %>%
  add_recipe(base_recipe) %>%
  add_model(glm_spec)

glm_fit <- fit(glm_workflow, data = train_df)
```

```
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```r
glm_pred <- predict(glm_fit, test_df, type = "prob") %>%
  bind_cols(predict(glm_fit, test_df)) %>%
  bind_cols(test_df %>% select(diagnosis))
```

```r
glm_class_metrics <- metric_set(accuracy, sens, spec)(
  glm_pred,
  truth = diagnosis,
  estimate = .pred_class
)

glm_auc <- roc_auc(glm_pred, truth = diagnosis, .pred_M)

glm_class_metrics
```

```
## # A tibble: 3 x 3
##   .metric  .estimator .estimate
##   <chr>    <chr>          <dbl>
## 1 accuracy binary         0.939
## 2 sens     binary         0.903
## 3 spec     binary         1
```

```r
glm_auc
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 roc_auc binary        0.0417
```

8.1 Compare models

```r
comparison_tbl <- bind_rows(
  lasso_class_metrics %>%
    select(.metric, .estimate) %>%
    mutate(model = "Lasso"),
  glm_class_metrics %>%
    select(.metric, .estimate) %>%
```

```r
    mutate(model = "Standard Logistic Regression"),
  lasso_auc %>%
    select(.metric, .estimate) %>%
    mutate(model = "Lasso"),
  glm_auc %>%
    select(.metric, .estimate) %>%
    mutate(model = "Standard Logistic Regression")
) %>%
  select(model, .metric, .estimate)

comparison_tbl
```

```
## # A tibble: 8 x 3
##   model                        .metric  .estimate
##   <chr>                        <chr>        <dbl>
## 1 Lasso                        accuracy  0.974
## 2 Lasso                        sens      0.958
## 3 Lasso                        spec      1
## 4 Standard Logistic Regression accuracy  0.939
## 5 Standard Logistic Regression sens      0.903
## 6 Standard Logistic Regression spec      1
## 7 Lasso                        roc_auc   0.000323
## 8 Standard Logistic Regression roc_auc   0.0417
```

9 Interpreting the Lasso Model

9.1 Coefficients

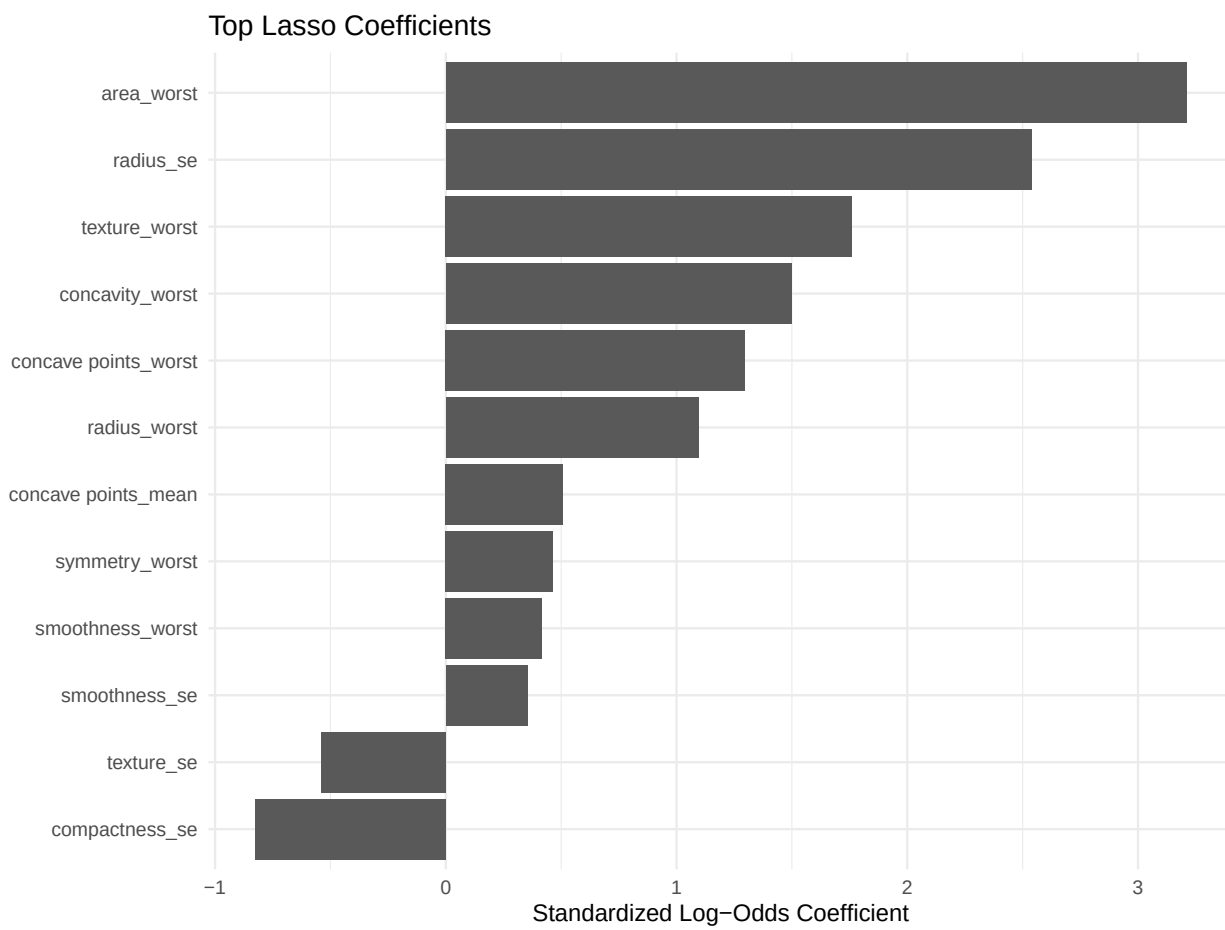
```r
coef_tbl <- tidy(extract_fit_parsnip(final_lasso_fit)) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    abs_estimate = abs(estimate),
    odds_ratio = exp(estimate)
  ) %>%
  arrange(desc(abs_estimate))

coef_tbl %>%
  select(term, estimate, odds_ratio, penalty, abs_estimate) %>%
  head(15)
```

```
## # A tibble: 15 x 5
##    term                   estimate odds_ratio penalty abs_estimate
##    <chr>                     <dbl>      <dbl>   <dbl>        <dbl>
##  1 area_worst               3.21       24.8   0.00191       3.21
##  2 radius_se                2.54       12.7   0.00191       2.54
##  3 texture_worst            1.76        5.81  0.00191       1.76
##  4 concavity_worst          1.50        4.47  0.00191       1.50
##  5 concave points_worst     1.30        3.66  0.00191       1.30
##  6 radius_worst             1.10        2.99  0.00191       1.10
##  7 compactness_se          -0.827       0.437 0.00191       0.827
```

```
##  8 texture_se              -0.539       0.583 0.00191       0.539
##  9 concave points_mean      0.507       1.66  0.00191       0.507
## 10 symmetry_worst           0.465       1.59  0.00191       0.465
## 11 smoothness_worst         0.417       1.52  0.00191       0.417
## 12 smoothness_se            0.353       1.42  0.00191       0.353
## 13 fractal_dimension_se    -0.218       0.804 0.00191       0.218
## 14 fractal_dimension_mean  -0.0912      0.913 0.00191       0.0912
## 15 texture_mean             0.0530      1.05  0.00191       0.0530
```

9.2 Plot top coefficients

```r
coef_tbl %>%
  slice_max(abs_estimate, n = 12) %>%
  mutate(term = fct_reorder(term, estimate)) %>%
  ggplot(aes(x = estimate, y = term)) +
  geom_col() +
  labs(
    title = "Top Lasso Coefficients",
    x = "Standardized Log-Odds Coefficient",
    y = NULL
  )
```



9.3 Interpretation

Because predictors were standardized before fitting, each coefficient reflects the change in the log-odds of malignancy associated with a one standard deviation increase in the predictor, holding the selected features constant. Positive coefficients indicate higher predicted odds of malignancy, while negative coefficients indicate lower predicted odds. Coefficients equal to zero were excluded by the lasso penalty, suggesting that they provided limited incremental information once correlated predictors were already in the model.

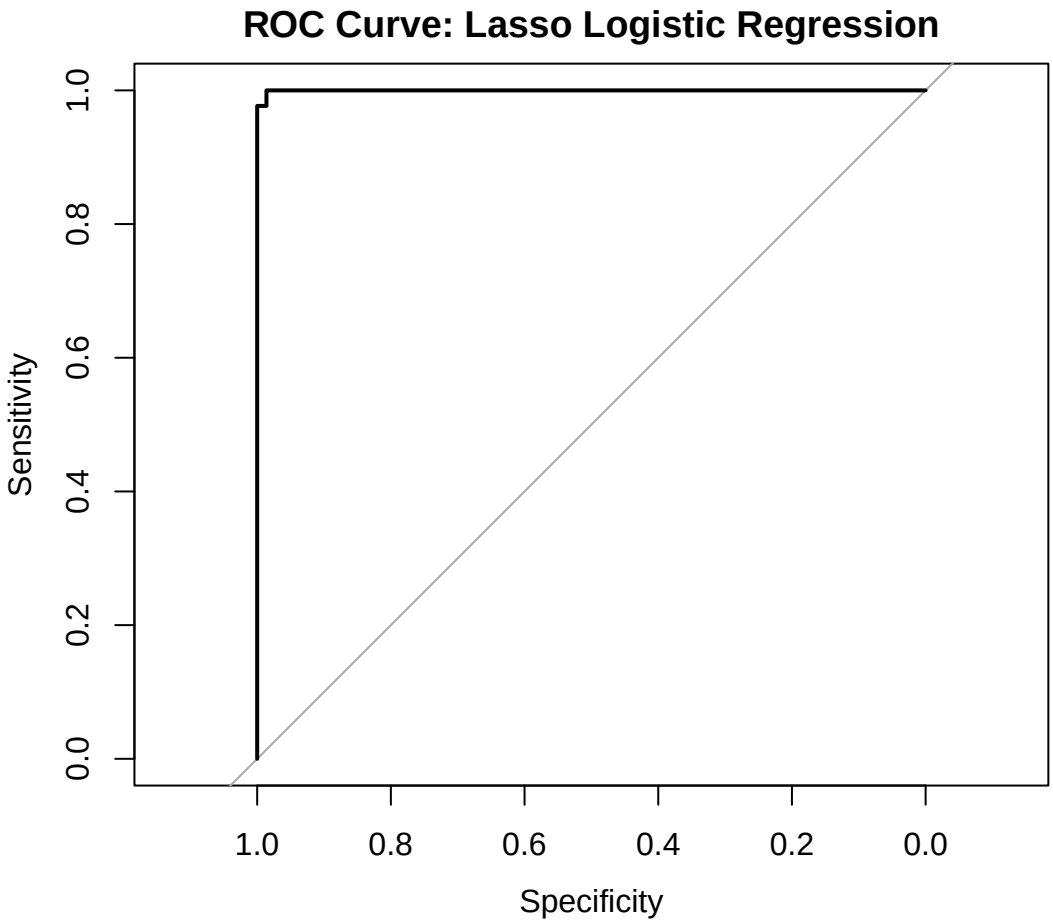
The largest positive coefficients are expected to correspond to measures of tumor size and boundary irregularity, such as worst-case radius, radius variability, and concave points. These findings are substantively plausible because malignant tumors tend to be larger and more morphologically irregular.

10 ROC Curve

```r
roc_obj <- roc(
  response = lasso_pred$diagnosis,
  predictor = lasso_pred$.pred_M,
  levels = c("B", "M")
)
```

```
## Setting direction: controls < cases
```

```r
plot(roc_obj, main = "ROC Curve: Lasso Logistic Regression")
```



11 Discussion

This analysis demonstrates that a regularized logistic regression can classify malignant versus benign tumors with strong predictive performance on a held-out test set. The comparison with an unpenalized logistic regression helps distinguish predictive strength from parsimony. Even if both models perform well, the lasso model is preferable when the goal is interpretability and reduction of redundancy among correlated predictors.

From a substantive perspective, the retained predictors are consistent with domain expectations: size-related variables and measures of boundary irregularity appear to be most informative for malignancy classification.

12 Limitations

This project should be interpreted as a predictive modeling demonstration rather than a clinical decision tool. The dataset is relatively small, drawn from a benchmark source, and does not by itself establish external validity. In addition, strong test performance on this dataset may partially reflect the clean structure of the benchmark rather than realistic clinical noise. Future work could examine calibration, compare more model classes, and test generalizability on external validation data.

13 Conclusion

This portfolio project shows a full reproducible workflow in R: data cleaning, exploratory analysis, train/test evaluation, regularized modeling, baseline comparison, and interpretation. The lasso framework is particularly useful here because it produces a sparse, interpretable model while preserving strong classification performance.